Applying Deep Learning Techniques to Diagnose Pneumonia Using Lung X-ray Images

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1 Abstract

Pneumonia is a major cause of death globally and presents significant challenges in medical diagnostics [1]. Advancements in artificial intelligence, specifically computer vision using deep learning, can offer assistance to medical professionals in diagnosing pneumonia efficiently and with increased accuracy. Analysis of the lungs using chest X-rays is essential to diagnose respiratory conditions like pneumonia. In this study, we introduce an open-source integration of a convolutional neural network (CNN) trained on an extensive, publicly available dataset of segmented lung X-ray images containing 35,000 images. This framework provides a diagnosis, a certainty metric, and a heatmap that highlights regions of interest that could aid health professionals in their diagnosis. This tool shows promise to be a reliable means to enhance diagnostic accuracy and efficiency in the healthcare industry.

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2 Introduction

By integrating machine learning in medical diagnostics, we have the potential to create a future where interpretations on radiograph images are accurate and reliable. Analyzing chest X-ray images with artificial intelligence has the power to improve treatment outcomes by providing medical professionals with complementary tools in their analysis of these images. By contributing to this field of research, we hope to propel medical diagnostics towards reliability and autonomy.

A pivotal role in diagnosis is played by radiology, which is a non-invasive procedure that provides a visual insight into the patient's internal state [2]. Among the most common radiology tests are chest X-rays, providing important visual information about the functionality of vital organs such as the heart and lungs. Chest X-rays are essential to the diagnosis of respiratory diseases such as lung cancer, pneumonia and tuberculosis [3][4].

Radiology reports are in high demand, however, there is a relative unavailability of radiologist specialists. The lack of specialists can lead to the risk of inaccurate diagnosis due to evaluations being given by non-specialists [5].

Deep learning, a subset of machine learning, has significant potential to resolve many of these issues, particularly through the use of convolutional neural networks. The application of deep learning in medical imaging is a widely researched area, with a strong emphasis on the uses of computer vision. Significant strides have been made, such as the introduction of U-net—a neural network that greatly improves biomedical imaging segmentation—and generative adversarial networks (GANs) for anomaly detection [10][11].

Studies like those by Racic et al. and Pant et al. tend to either excel in accuracy but lack a large dataset, or lack transparency in datasets and code. While such studies are impressive and critical to our work, they are often irreproducible or lack generalizability due to small datasets and overfitting. Our goal is to achieve the best possible accuracy on a large, free, and public dataset using a pre-trained CNN, while also providing insight into the model's decision making.

3 Related Work and Background

Research on creating sophisticated algorithms for medical diagnosis is an up-and-coming field that can be highly beneficial to health professionals worldwide. Providing large datasets of chest X-rays to sophisticated machine learning models presents the potential for AI to understand the subtleties of radiographic images, efficiently diagnosing pneumonia. This research field is growing with the ever-expanding possibilities of deep learning models. Large amounts of critical care information, including lung X-ray images, were provided via the MIMIC-IV database shared on PhysioNet. This information is imperative to creating reliable deep learning diagnostic models. The MIMIC-CXR dataset, also found on PhysioNet, provides a large collection of clinical reports that correspond to chest X-rays; this is very necessary for both training and testing deep models. The MIMIC-CXR-JPG dataset contains 243,324 frontal chest X-ray images containing images of lung X-rays segmented by the SA-UNet 13 architecture.

Our convolutional neural network (CNN) architecture was trained on labeled segmented lung X-ray images with different abnormalities from the CXLSeg dataset, which contains segmented chest X-ray radiographs based on the MIMIC-CXR dataset. The results by Yifan Zhang [5], published by Northwestern University, present the performance of CNNs in recognizing pneumonia specifically in the MIMIC-CXR-JPG dataset and thus set a benchmark for our model.

Recent deep learning developments, such as those by Zhang et al. (2020), where it is shown that CNNs can be very effective at differentiating between pneumonia and healthy lung X-rays, also help inform our methodology.

We have used Gradient-weighted Class Activation Mapping, a technique described by Selvaraju et al. (2017), to provide interpretability in the convolutional neural network's decision-making. Grad-CAM increases the transparency of our model's inner workings by generating a heatmap that points out areas in images that contribute most to the prediction made by the CNN. Caruana et al. (2015) consider model interpretability critical in clinical settings and promote techniques that provide insight into the reasoning behind AI predictions.

Comparative research into deep learning architectures by Rajpurkar et al. (2018) has been very instrumental in helping us fine-tune our CNN architecture for pneumonia diagnosis. Finally, studies like those by Lakhani and Sundaram (2017) on the challenges and progress in machine learning-aided diagnosis of pneumonia underpin the necessity for better diagnostic tools, highlighting the applicability and potential benefits of an AI-based approach. Taken together, these references help in developing and refining our diagnostic tool and ensure its utility to the medical practitioner.

4 Data and Methodologies

4.1 Dataset and Preprocessing

For this research, we used the Chest X-ray with Lung Segmentation (CXLSeg) dataset (see Fig 1 for a sample batch). The CXLSeg dataset is a large, freely accessible collection of segmented chest radiograph images created to bolster medical image segmentation research. It was derived from the MIMIC-CXR dataset and includes 243,324 frontal-view chest X-ray images. Each of these X-ray images was preprocessed with AI-generated segmentation masks that highlight the lung regions. These masks are essential for fine-tuning AI-based diagnostic systems by enabling models to focus on the most relevant parts of an image, thereby improving the accuracy of medical image analysis. The CXLSeg dataset is applicable to a broad spectrum of computer vision tasks, including medical image classification, semantic segmentation, and the creation of medical reports. Specifically, models trained with CXLSeg have proven to be quite effective. For example, the SA-UNet model achieved an IoU score of 91.97% and a Dice similarity coefficient of 96.80% for lung segmentation [12].

As the dataset was large, we decided to split it into three sections, each containing 81,110 chest X-ray images. This allowed us to shorten the amount of time needed for the preprocessing. The biggest issue with our current dataset was the fact that lung segmentation was carried out by an AI, which resulted in about 10% of the images being of poor quality with incorrect segmentations. This forced us to manually remove the X-ray images with low quality (see Fig 1 for Good and Poor Quality Segmentation) and incorrect segmentation, leaving us with 220,203 images. Furthermore, we also needed to rotate the images as they all came oriented differently.

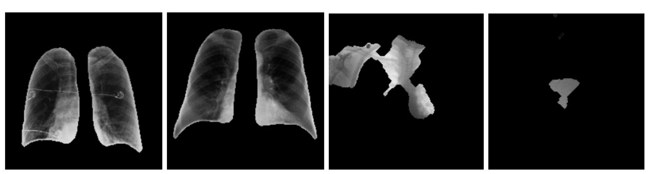

Fig. 1 Chest X-ray Images from the Dataset. This shows the masking of the lungs which was given in the dataset. The two images on the left show good segmentation, while the two images on the right show poor segmentations.

4.2 Labeling

Additionally, the original CXLSeg dataset did not differentiate between pneumonia and non-pneumonia patients. It contained X-ray images from over 10 different respiratory diseases such as COVID-19 and Pneumothorax. The dataset included X-ray images that were not labeled, which made it so we had to label the images accordingly. The dataset contained a metadata file that included values of [-1, 0, 1] under each corresponding column for each disease. Each of these integer values had its own meaning when it came to the diagnosis. A -1 indicated that the findings were uncertain or not readable, a 0 indicated there was no instance of the disease in the lung X-ray, and a 1 indicated there was an instance of the disease in the lung X-rays.

From this, we were able to create a Python program that looked for only instances of a 1.0 under the "Pneumonia" and "No Finding" columns and placed those X-rays in their respective folders, either pneumonia or normal. As our framework only dealt with pneumonia and non-pneumonia patients it was unnecessary for us to look at other diseases or X-rays related to other diseases. With this, we not only condensed the full dataset down to about 165,000 X-ray images but we made sure that our dataset was properly cleaned and only included relevant information.

We needed to split our dataset into testing, validation, and training folders. This was so that rather than the AI using the full dataset and getting overlaps between X-ray images, we could ensure that 80% of our dataset was being used for training and was unique compared to the 10% used for validation and 10% for testing. This percentage was recommended from the studies that used MIMIC. This was simply done using Python code that went through our full dataset and randomly split the data into its corresponding folders.

4.3 Model development

Our research for finding the most effective and accurate AI model for pneumonia classifications with the CXLSeg dataset was not as fruitful as we hoped. There had been research done using so many models with similar accuracy rates that we felt deciding on one particular model would limit our ability to potentially improve results. We aimed to try most of the models and test them – models such as ResNet50, ResNet34, Vgg16, and Nicholas Muchinguri's fine-tuned model for pneumonia [15] – without missing anything that might have seemed to hold some potential. In so doing, we were trying to find a model that provided the best trade-off between model accuracy, computational efficiency, and good generalizability across different datasets. This rigorous approach further enabled us to gain a finer understanding of the advantages and disadvantages of each type, which will help in coming up with some solid, well-informed solutions for pneumonia classification.

We chose to adopt a transfer learning approach, where these models were trained on our CXLSeg dataset after being pre-trained on large datasets such as ImageNet. Transfer learning speeds up the training phase and enhances performance by letting the models exploit features already learned, possibly useful for the task at hand.

First, we decided to implement our VGG16 model using TensorFlow Keras on Khan's Kaggle Dataset, which was much smaller in size. We used this model due to its comprehensive set of tools and high-level interface, that makes training and prototyping rather easy. In the process of development, though, we did face convergence problems of the models and the difficulty in tuning hyperparameters with a larger dataset, which turned out to bring very underwhelming performance with long training times. While with a smaller dataset, we were able to get much higher accuracy this was not the case when we ran the framework with the much larger CXLSeg dataset.

We worked on a reduced dataset of 35,000 images because of our limited computing resources. This was done on Kaggle, as it is a free cloud server to run machine learning models. In this way, we could run several iterations which wouldn't put too much stress on computing. Further, for testing the hyperparameters like learning rates and early stopping conditions, we also created a small subset dataset with 720 images for each folder, leading to a total of 2,160 X-rays. Early stopping was very useful in avoiding overfitting because it allowed us to terminate training at exactly the right time–that is when the model's performance on the validation set began to stabilize.

We decided to use ResNet34 for our smaller dataset, knowing that the design of the model is lighter and that it's easier to test with it, along with its smaller number of layers. Knowing that we had a larger dataset, we wanted to go with ResNet50, hoping that a deeper design would capture more complex features and result in better performance. Using ResNet34 with this smaller dataset, the model failed to generalize as much. With ResNet50, however, while having increased performance, also came issues associated with increased training time and thus computational load, which made hyperparameter tuning very hard and stable convergence even harder. These challenges underscored the importance of careful balancing between model complexity, dataset size, and computational resources, and drove us back to revisit the approach with further optimization strategies.

Knowing these weaknesses, we decided to look at other frameworks and changed to Python with PyTorch for Fast.ai. Using fastai gave us more flexibility and efficiency, enabling us to experiment on different models and methodologies. We used the fastai on Python module with PyTorch to accelerate the process since it had a much more user-friendly implementation process, particularly in regard to charting training progress and learning rates. Also, given our computing limitations, fastai was an efficient way to experiment with different models and hyperparameters.

Explainability was an important consideration in our effort to develop a reliable diagnostic tool for medical practitioners. We did this by incorporating Grad-CAM, or gradient-weighted class activation mapping, into the framework. Grad-CAM creates a heatmap that indicates which areas of the picture most impacted the model's conclusion. Grad-CAM works by using the gradients of the output class with respect to feature maps of a convolutional layer. It can also be used in identifying overfit models or otherwise not generalizing, focusing on the wrong parts of an image. This improves the utility of the tool to a medical professional who must understand the logic behind the AI outputs in addition to adding another layer of validation to the predictions of the AI.

Knowing the ”valley” and ” minimum point on steepest drop (Slice(min, steep))” values on the learning rate curve helps determine the learning rate. Minimum point on steepest drop is the steepest decrease in loss; hence, it's the most efficient learning rate. The valley marks the point where the loss is reduced without the danger of overfitting. We are able to optimize our model for improved performance by understanding these values.

5 Results

The source code used to generate these results is available at:
https://www.kaggle.com/code/riyaanjain/pneumonia-detector
A total of 35,000 chest X-ray images were used to train, test, and validate the ResNet50 model. Due to the vastness of our dataset and the lack of computation power, we randomly selected 2,160 images from the dataset, split those into 720 images for each of the train, test and validation sets, and used it to find the optimal conditions for high accuracy. This allowed us to fine-tune and adjust the hyperparameters more easily, eventually transferring our findings on the small dataset to the final implementation for the large dataset. We used ResNet34 for the smaller dataset.

Though attempts were made to incorporate Nicholas Muchinguri's fine-tuned model specifically for pneumonia detection, it proved to be too resource intensive, taking considerably longer to run than ResNet. VGG16 was also explored on our discontinued implementation that used TensorFlow, however, accuracy was inferior on the CXLSeg dataset, though was slightly better on Khan's dataset found on Kaggle, boasting 97% accuracy, as compared to 96% achieved with ResNet50. The following section will discuss the results of running ResNet34 on the smaller dataset.

To begin with, we ran the model on the smaller dataset to find the optimal learning rate without any fine-tuning. We obtained the plot in Figure 2 for learning rate against loss, and trained the model using both the learning rates of the valley value and the learning rate where the validation loss decreases most sharply and is at its minimum. After training the model for 4 epochs, we compared the accuracy and loss values obtained from training the model using the different learning rates, displayed in Table 1 and Table 2.

When initially training the smaller dataset, when the learning rate is set to a narrow region around the minimum loss on the steepest drop (Table 3), the final accuracy is 4.31% higher than when the learning rate is set to the valley value, however, the validation and training losses are also higher. The longest time to run an epoch using the learning rate of the minimum loss on the steepest drop was 9.05 minutes, as compared to 3.52 minutes for the valley learning rate. Without fine-tuning, we achieved an accuracy on the test dataset of 52.92% when training the model with the smaller dataset. When testing the model with an image from the validation set of a pneumonia patient, it correctly predicted the presence of pneumonia with 71.63% certainty.

Moving forward with the valley learning rate for all further runs, we trained the model on the smaller dataset and added fine-tuning. The initial run of fine-tuning spanned 20 epochs, but with the addition of early stopping with a patience of 3 epochs, the fine-tuning stopped after the 7th epoch, indicating no improvement since the 4th epoch in validation loss, presented in Table 3.

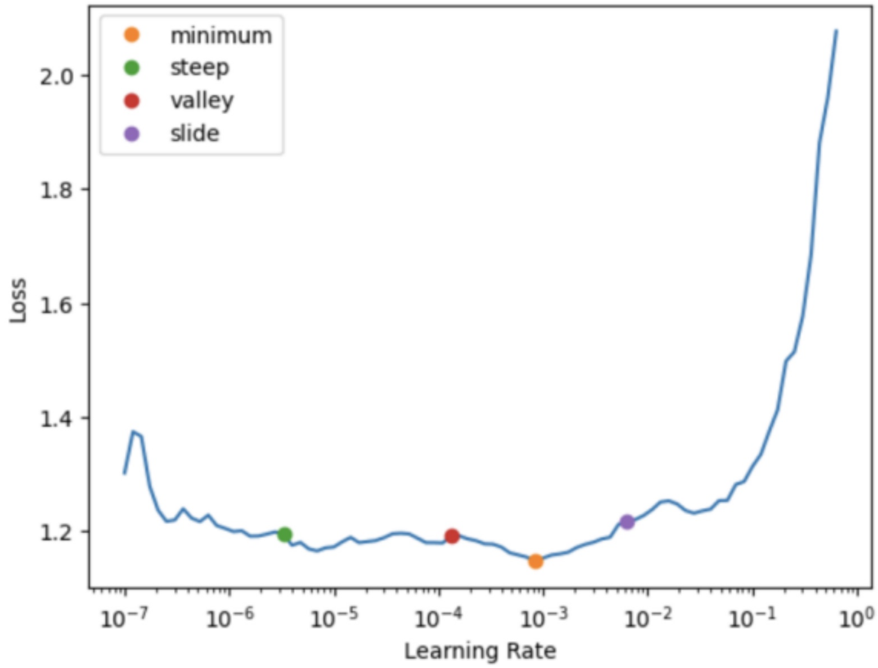

Fig. 2 Learning Rate vs Loss plot. This plot highlights key learning rate values with coloured dots, with a corresponding key on the top left.

epoch	train_loss	valid_loss	Accuracy	Time
0	1.101559	0.883700	0.525000	03:52
1	1.154545	0.823594	0.526389	03:45
2	1.134318	0.845187	0.511111	03:49
3	1.108461	0.881691	0.523611	03:50

Table 1: Loss and accuracy results from training the model on the smaller dataset using the valley learning rate value.

epoch	train_loss	valid_loss	accuracy	time
0	1.099027	0.860371	0.562500	08:27
1	1.152470	0.911116	0.523611	09:05
2	1.119738	0.968770	0.550000	08:52
3	1.126968	0.850834	0.566667	08:33

Table 2: Loss and accuracy results from training the model on the smaller dataset using the narrow region around the minimum loss on the steepest drop as the learning rate.

epoch	train_loss	valid_loss	accuracy	precision_score	recall_score	f1_score	time
0	1.069172	0.871970	0.573611	0.575931	0.558333	0.566996	02:37
1	1.043682	0.871399	0.587500	0.581818	0.622222	0.601342	03:13
2	1.058284	0.891779	0.595833	0.583133	0.672222	0.624516	04:16
3	0.991160	0.844746	0.595833	0.591512	0.619444	0.605156	04:16
4	0.988348	0.894822	0.544444	0.541885	0.575000	0.557951	03:23
5	0.967968	0.904318	0.590278	0.577197	0.675000	0.622279	04:16
6	0.943897	0.923989	0.584722	0.570115	0.688889	0.623899	04:21

Table 3: Loss, accuracy, precision, recall and F1 scores presented per epoch when fine-tuning the model on the smaller dataset using the valley learning rate.

After optimizing hyperparameters by unfreezing the model and finding a new learning rate, the valley learning rate value decreased from slightly less than 0.001 to between 0.0001 and 0.00001. The model is then run to fine-tune for 10 epochs, but stopped at epoch 5 due to early stopping, indicating no improvement since epoch 2, evident in Table 4.

epoch	train_loss	valid_loss	accuracy	precision_score	recall_score	f1_score	time
0	0.847084	0.913624	0.570833	0.559441	0.666667	0.608365	03:50
1	0.834768	0.890128	0.572222	0.564039	0.636111	0.597911	04:05
2	0.832446	0.866220	0.591667	0.600610	0.547222	0.572674	03:31
3	0.866056	0.910678	0.583333	0.571090	0.669444	0.616368	03:32
4	0.863313	0.884044	0.590278	0.583120	0.633333	0.607190	03:42
5	0.846623	0.894150	0.577778	0.565728	0.669444	0.613232	03:32

Table 4: Loss, accuracy, precision, recall and F1 scores presented per epoch when fine-tuning over 10 epochs after optimizing hyperparameters for the smaller dataset.

Figure 3 shows that there is a general negative trend in loss. After fine-tuning and training with the small dataset, the total accuracy on the test portion of the dataset is 59.03%, a 6.11% higher accuracy than training without fine-tuning. The same pneumonia X-ray image was used to test the model as when no fine-tuning was conducted, and the model correctly predicted pneumonia with certainty of 83.24%. The Grad-CAM was also run and produced the heatmap in Figure 4, highlighting the regions of interest in the image that the CNN used most to determine its decision. The following section discusses the full, 35,000 image dataset, which was split into 80:10:10 (%) for train, test, and validation respectively, as suggested by the Chest X-ray with Lung Segmentation dataset. When running our code on the full dataset, we discovered that each epoch took more than 2 hours to complete (Table 5). The initial training run, since it was 4 epochs, took around 9 hours to run. After running

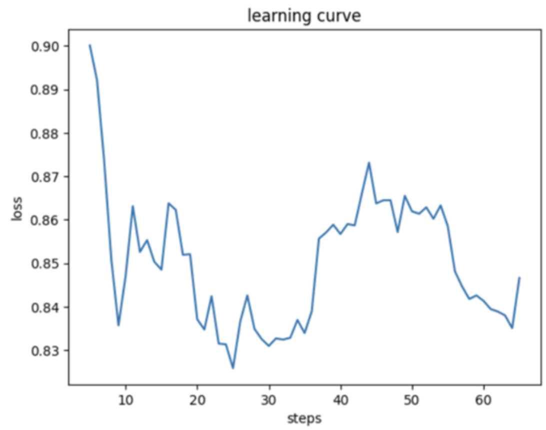

Figure 3: Learning curve plot with steps on the x-axis and loss on the y-axis.

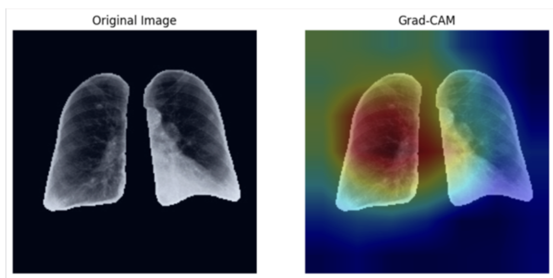

Figure 4: Heatmap produced by Grad-CAM, highlighting regions of interest to the CNN when trained on the smaller dataset.

the cell to find the learning rates, shown in Figure 5, and conduct initial training, our session timed out due to Kaggle's 12 hour session limit restriction. Due to the Kaggle timeout restrictions, we had to reduce the number of epochs for initial training down to 2 (see Table 6), and removed all 20 epochs of initial fine-tuning. The hyperparameter optimization cell was run accidentally and could not be canceled without disrupting the session, and we were only able to fine-tune for 1 epoch after that. The learning curve displayed a stronger negative correlation on the full dataset than the smaller dataset, as seen in Figure 6. Figure 7 shows the confusion matrix of the run on the full dataset. A 56.96% test accuracy was reported. No test image was able to process due to a human error in file path, and once the error was spotted the session had already timed out.

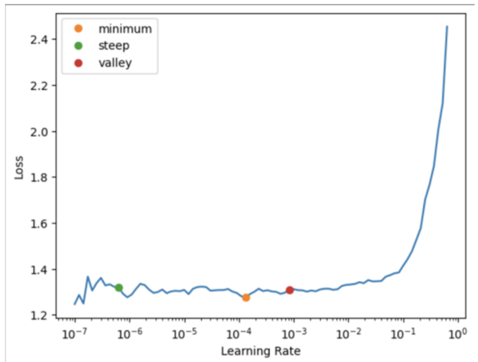

Figure 5: Learning rate vs loss for full dataset.

epoch	train_loss	valid_loss	accuracy	precision_score	recall_score	f1_score	time
0	0.813468	0.691017	0.563733	0.558550	0.608000	0.582227	2:05:08
1	0.688426	0.679791	0.549067	0.593117	0.312533	0.409361	2:16:55
2	0.677269	0.667919	0.585333	0.591429	0.552000	0.571034	2:17:46
3	0.661245	0.667244	0.572267	0.595355	0.451200	0.513350	2:20:23

Table 5: Metrics when conducting initial training on the full dataset for 4 epochs.

epoch	train_loss	valid_loss	accuracy	precision_score	recall_score	f1_score	time
0	1.109331	0.795858	0.550667	0.568741	0.419200	0.482653	2:08:12
1	0.966887	0.749632	0.564267	0.589858	0.421867	0.491915	2:11:44

Table 6: Metrics for initial training on the full dataset for 2 epochs.

epoch	train_loss	valid_loss	accuracy	precision_score	recall_score	f1_score	time
0	0.902063	0.706253	0.574667	0.582742	0.525867	0.552846	3:12:02

Table 7: Metrics when fine-tuning for 1 epoch on full dataset.

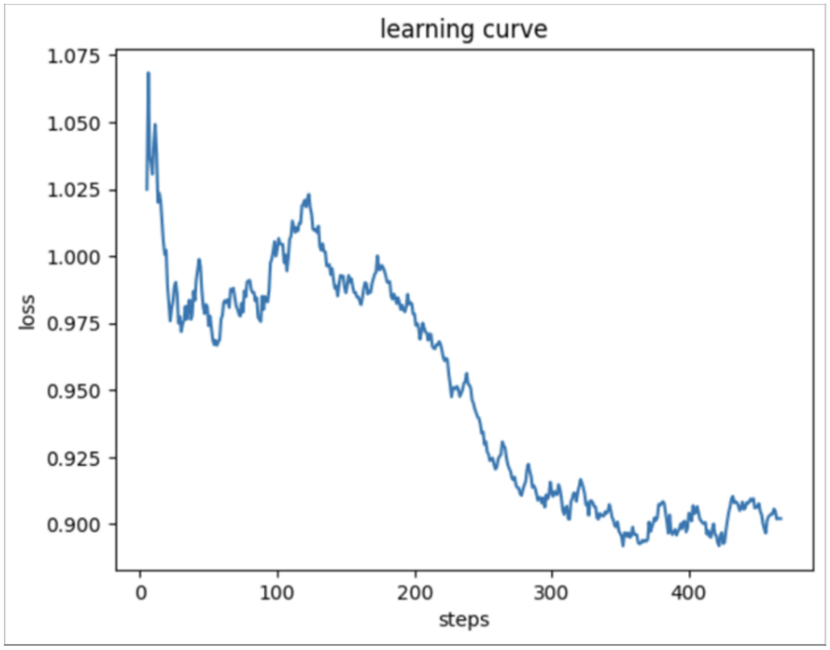

Figure 6: Learning curve: steps vs loss plot on the full dataset.

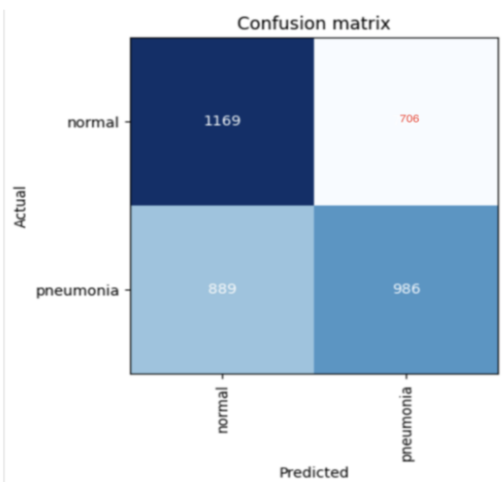

Figure 7: Confusion matrix of final result

6 Conclusion

Our research aims to build a framework to use deep learning models to efficiently diagnose pneumonia. Khan's dataset on Kaggle was considerably smaller in size with very clear and consistent images compared to CXLSeg, which led to us achieving much higher training accuracies than what we ended up with for CXLSeg. However, our values indicated strong overfitting and a lack of generalizability, underperforming on unseen data. Khan's dataset was also free of any foreign artifacts, such as pacemakers, a common occurrence found in the CXLSeg images. Our testing with the smaller dataset (2,160 X-rays) allowed us to find the optimal learning rate, which we found, accuracy-wise, to be a narrow region around the minimum point on the steepest drop of the learning rate vs. loss plot. We also experimented with the valley value for the learning rate, and while we found lower accuracy, the lower loss and significantly shorter time per epoch were essential in our ability to run the model on our full dataset.

Fine-tuning the model significantly improved the accuracy of our results with the smaller dataset. Implementing early stopping was also pivotal to the success of this, as not only did it save time, but it prevented overfitting by monitoring the validation loss and making sure the model continues to prioritise generalizability. Table 3 shows an uncertain precision score progression, meaning the model is not getting any better or worse at correctly identifying positive instances among all predicted positives. The recall score shows a steady increase, suggesting the model becomes better at identifying true positives. The F1 score, being a harmonic mean of precision and recall, increases through the epochs, implying a good trade-off between them. After hyperparameter optimization, the learning rate decreased, allowing more gradual updates to the weights and helping prevent overfitting by learning more subtleties in the images. The recall, precision, and F1 scores all hold up relative to pre-optimization. The continued negative trend in loss is also great at proving the lack of overfitting and generalization of the model. The test accuracy with fine-tuning was significantly better, proving the importance of fine-tuning. The learning curve's general negative trend is also a good sign of the model getting better for unseen data, and the heatmap produced by the Grad-CAM correctly focuses on the fluid and pus buildup in the left lung.

The full dataset that we used for our model was very large and contained lung segmentation done by the SA-Unet model. Unfortunately, this led to a significant number of images being segmented incorrectly and some of them even being completely blank. Though we cleaned as much of the dataset as we could manually, due to the sheer number of images, a lot of incomplete segmentations remained in the data, affecting the model's accuracy. The 12 hour Kaggle session timeout restriction was also a major hurdle we never fully overcame. We were unable to apply most of the changes we implemented that we learnt from training on the small dataset onto our full dataset. Due to a lack of powerful computers between either of us, we were unable to find an alternative to this. During our initial training run with the full dataset, where we ran 4 epochs, the accuracy was already very promising, and the decrease in loss was sharp (Table 5). The precision increased, the recall decreased, and the F1 also generally decreased, but generally fluctuated. With these initial training results, we believe that, if we were able to fine-tune and run the exact code as we did for the smaller dataset instead of having to decrease the number of epochs, our accuracy would've been comparable to that of Zhang (2024), where he achieved around 70% accuracy using various methods like downsampling and upsampling on the MIMIC-CXR-JPG dataset.

Instead, we were limited to 2 epochs to train, and 1 epoch for fine-tuning, which was very underwhelming but still produced a very strong negative learning curve, showing great promise in the ability of CNNs to benefit from and generalize well with big datasets of chest X-ray images. An accuracy of 56.96% test accuracy was surprisingly good for such few epochs to fine-tune the model, greatly outperforming the smaller dataset when it was not fine-tuned. The confusion matrix indicates that the model was strong at detecting true positives for 'normal' cases, which indicated non-pneumonia, but still struggles with determining whether images of lungs impacted by pneumonia are infected or not. By integrating Grad-CAM to produce heatmaps of the regions of interest to the CNN, we provided interpretability to our results in order to maximize our tool's applicability in the medical workspace.

Pneumonia can be classified into four categories: bacterial, viral, fungal and aspiration. Future work in this research field can help integrate the evaluation of these specific types of pneumonia in order to excel clinical diagnostics even further. Classification of other respiratory diseases such as tuberculosis and COPD could also be integrated into the same framework in the future, helping expand the usability of frameworks like ours. Through this study, we aim to contribute to the progression of artificial intelligence in medical diagnostics, which has the potential to produce accurate, efficient, and affordable interpretations of radiograph images, bringing a positive change to the medical industry.

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